# Ravinder

Updated: October, 2026

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## Visiting Fellowships

- Visiting Scholar, Indira Gandhi Institute of Development Research (IGIDR), Mumbai, India. September-December 2025
- Visiting Ph.D. Fellow, United Nations University World Institute for Development Economics Research (UNU-WIDER), Helsinki, Finland. September-November 2022

## Education

- PhD in Quantitative Economics, Indian Statistical Institute, Delhi 2020 - 2026
  (Advisor: Prof. Abhiroop Mukhopadhyay) [Start: Dec 20; Submission: Nov 26; Defence: Mar 27 (Exp.)]
- Master in Science in Quantitative Economics (MSQE), Indian Statistical Institute, Delhi 2018 - 2020
- Bachelor of Arts (honors) Economics, Kirori Mal College, Delhi University 2015 - 2018

## Research Interests

Human Capital, Development Macroeconomics, Economics of Education, Labour Economics, and Rural Economy.

## Working Paper

- **Castelló-Climent, Amparo, Abhiroop Mukhopadhyay, and Ravinder. Transforming rural economies through tertiary education: Evidence from India. No. 2023/89. WIDER Working Paper, 2023.**
  Abstract: This paper examines whether tertiary-educated individuals who live in rural India contribute to local prosperity. Using village-level data, we estimate the causal impact of tertiary education on rural development, exploiting the location of Catholic missions circa 1911, historically associated with the expansion of higher education, as an instrumental variable. A one percentage point increase in the village share of tertiary-educated individuals raises per capita consumption by about 7.2 percent, a result robust to using night-light density as an alternative prosperity measure and to a wide range of further checks. We provide evidence consistent with two mechanisms: a conventional channel, in which tertiary-educated individuals access productive private-sector employment locally and in nearby towns, and an agricultural channel, in which households with a tertiary-educated member show higher agricultural productivity, greater access to technical advice, and more crop diversification. The findings point to a distinct form of structural transformation occurring within rural areas, alongside that driven by migration to cities.

- **Foundational Skills and Labor Market Outcomes: Evidence from India**
  Abstract: The expansion of primary schooling in developing countries has not been matched by gains in learning. This paper asks whether a program that raised foundational learning in primary school also raised earnings in early adulthood. I exploit the eligibility rule of India's District Primary Education Programme (DPEP), which targeted districts whose 1991 female literacy rate fell below the national average, in a regression discontinuity design. I observe the same birth cohorts in rural India at two stages of life: as children aged 8-11 in the Annual Status of Education Report from 2006 to 2011, and as young adults aged 22-25 in the Periodic Labour Force Survey from 2017 to 2022. DPEP eligibility raised test scores by 0.14 standard deviations, with similar gains for boys and girls. As young adults, the same cohorts earn about 32 percent more per week in regular wage or salaried jobs, but I find no earnings gains among the self-employed or casual laborers. The earnings gain is positive and significant for men but not for women. The results are robust to a range of checks. More broadly, the findings suggest that the returns to education in developing countries depend not only on how long children stay in school but also, importantly, on what they learn there.

- **Relaxing Entry Constraints: College Expansion and Tertiary Attainment in Rural India**
  Abstract: Tertiary education attainment among young adults in India has risen sharply over the past three decades, alongside a large expansion in the number of higher education institutions. This paper asks whether expanded access to colleges raised local rural tertiary education attainment. I study Rajasthan, which follows a similar trend to India in rural tertiary attainment but has experienced the highest rate of college expansion among Indian states. I combine geocoded college locations from the All India Survey on Higher Education with unique cohort-level records from the Socio-Economic and Caste Census 2011. Using individuals aged 22 to 31 in 2011, I construct a village-by-birth-cohort panel spanning 1998 to 2007, linking each cohort to the college supply prevailing when it reached college-entry age. The expansion began in 2003, and I exploit the staggered entry of colleges across villages. Exposure to a college within 15 km raised the tertiary attainment of exposed cohorts by about 1.4 percentage points two years after exposure and by 2.2-2.4 percentage points after four to five years, with no evidence of pre-trends. The effect is robust to several checks, including alternative comparison groups, distance thresholds, restricted cohort window, and adjustments for local economic activity, and holds for both men and women. Turning to college placement, district land fractionalization in 1999 is strongly associated with subsequent college growth. More- and less-fractionalized districts had similar college stocks before the expansion but diverged only after it began. This pattern is consistent with fragmented, lower-value agricultural land offering cheaper sites for a land-intensive, largely private expansion of higher education.

# Work in Progress

- **Beyond Cities: Are Towns the New Engines of Growth in India?**

# Master's Thesis

- **Human Capital Inequality and Rural Income - The case of Rural India.**
  Abstract: This paper estimates the impact of human capital inequality on rural income for 27 states of India at the village level, taking advantage of Census (2011) and Socio-Economic and Caste Census (2011) data. To address the endogeneity of human capital inequality and mean education, we use the distance to the nearest Catholic and Protestant missionaries (settled in the early 20th century) as an instrument. Our instrumental variable results find a positive

effect of human capital inequality and mean education on mean income, while the interaction of these variables shows a negative effect on mean income. The IV results estimate that one standard deviation increase in human capital inequality increases mean income by 221%, and one year increase in mean education increases mean income by 136%. The results are robust to alternative measures of income, such as nightlight intensity. The possible mechanism to explain this relationship is through the occupation structure. Our results are robust and significant at the 1% level.

# Other Writing

- **Predicting School Education Outcomes in India: A Machine Learning Approach (with Priya Bareja).**
  Abstract: Optimal policy design is a three-step approach —qualitatively capturing various dependencies relating input(s) to output(s), converting these qualitative insights into quantitative relationships that predict the output for various combinations of the inputs and finally, using this quantitative relationship to design optimal input combinations. This paper estimates a prediction model to predict the school learning outcomes for a given set of inputs like teacher characteristics, school infrastructure and health indicators of children. We merge district-level data from four datasets on the National Data and Analytics Platform (NDAP) to get the features and target variables, and train sixteen machine learning algorithms on the data. We use K-fold cross validation and choose the algorithm with the least test error —the Extra Trees Regressor. Thus, learning outcomes produced with a given input combination can be predicted sufficiently accurately by the policymaker.

- **Higher Education and its Impact on the Non-Farm Activities in Rural India (with Abhiroop Mukhopadhyay).**
  Abstract: This paper investigates the causal impact of tertiary education on the rural economy of India through entrepreneurship and jobs in the non-farm sector. Leveraging village-level data from the Economic Census 2013 and the Socio-Economic and Caste Census 2011, we construct a measure of the tertiary-educated population and examine its relationship with non-farm economic activity. Our results show that a higher share of tertiary-educated individuals significantly increases the number of non-farm enterprises, entrepreneurs, and overall employment in the rural non-farm sector. Further, it increases the probability of private salaried employment that involve commuting. To address endogeneity concerns, we use the historical presence of Catholic missions as an instrument for tertiary education. The findings contribute to the broader literature on human capital and economic development by highlighting the role of education in shaping the rural economy.

# Award, fellowship and Grants

- Best Paper award at 7th Annual Economics Conference, Amrut Mody School of Management, Ahmedabad University. 2026
- Visiting Ph.D. fellowship, IGIDR, Mumbai, India. 2025
- Awarded with an OSF Grant to Attend the XXXI meeting of the Economics of Education Association. 2023
- Project (STEM Education and Economic Growth) Grant by Planning and Policy Research Unit (PPRU). 2023
- Visiting Ph.D. fellowship, UNU-WIDER, Helsinki, Finland. 2022

- First prize (with Priya Bareja) for best research paper and visualization using NDAP data organized by Niti Aayog, Govt of India. 2022
- Senior Research Fellowship, Indian Statistical Institute, Delhi. 2022 - ongoing
- Junior Research Fellowship, Indian Statistical Institute, Delhi. 2020 - 2022
- Masters Fellowship, Indian Statistical Institute, Delhi. 2018 - 2020

## Conference and Workshop

- Poster Presentation at WIDER Development Conference, Green Industrialization and Inclusive Growth in a Fractured World Order, ISID, New Delhi. 18-20 March 2026
- Presented at Economics Research Conference (ERC) 2026, Indian Institute of Management Calcutta. 13-14 March 2026
- Presented at 5th Society for Economics Research in India Doctoral Conference (SERI-D) 2026, Amrut Mody School of Management, Ahmedabad University. 06-07 March 2026
- Presented at Research Scholars' Day 2026, Department of Economic Sciences, Indian Institute of Technology Kanpur. 13-14 February 2026
- Presented at 7th Annual Economics Conference, Amrut Mody School of Management, Ahmedabad University. 09-10 January 2026
- Teaching Assistant and Attendee of Workshop on Causal Identification: Rethinking Econometric Methods at Saint Xavier's College, Jaipur. 27-31 December 2025
- Presented at Annual Conference on Economics and Public Policy, O.P. Jindal Global University, Sonipat. 22-23 December 2025
- Teaching Assistant and Attendee of Workshop on Basics of Statistics & Econometric Methods at the Indian Institute of Public Health, Shillong. 24-28 November 2025
- Presented at Symposium on Gender Inequality in Indian Higher Education, Visva-Bharati University, Santiniketan. 04 November 2025
- Presented at 19th Annual Conference on Economic Growth and Development, Indian Statistical Institute Delhi. 19-21 December 2024
- Teaching Assistant and Attendee of Workshop on Data Analytics and Causal Design at the School of Economics at University of Hyderabad. 16-18 August 2024
- Presented at Recent Developments in Economics Research: Theory and Evidence, a conference hosted by Centre for International Trade and Development, Jawaharlal Nehru University, New Delhi. 07-08 March 2024
- Presented at the Meeting of Young Minds in Frontiers of Economics, a conference hosted by INET-YSI at Indian Institute of Technology Bombay. 20-22 February 2024
- Presented at the Winter School hosted by Delhi School of Economics, University of Delhi. 14-16 December 2023
- Presented at the XXXI Economics of Education Association meeting hosted by Santiago de Compostela University. 29-30 June 2023

- Attendee of the RISE Research Conference on Education Economics, Indian Institute of Management Ahmedabad (IIMA). 24-25 February 2023
- Presented at 17th Annual Conference on Economic Growth and Development, Indian Statistical Institute, Delhi. 19-21 December 2022
- Presented in the Advanced Graduate Workshop on Poverty, Development and Globalization, Azim Premji University, Bangalore. 10-23 July 2022
- Teaching Assistant and Attendee of Political Economy Theory and Data Analysis Workshop at GIPE, Pune. 09-11 June 2022

# Teaching

- Teaching Assistant for Econometric Method 1 January 2026 - July 2026
- Teaching Assistant for Workshop on Causal Identification: Rethinking Econometric Methods at Saint Xavier's College, Jaipur. 27-31 December 2025
- Teaching Assistant for Workshop on Basics of Statistics & Econometric Methods at the Indian Institute of Public Health, Shillong. 24-28 November 2025
- Teaching Assistant for Econometric Method 1 January 2025 - July 2025
- Teaching Assistant for Workshop on Data Analytics and Causal Design at the School of Economics at University of Hyderabad. 16-18 August 2024
- Teaching Assistant for Econometric Method 1 January 2024 - July 2024
- Teaching Assistant for Econometric Method 1 January 2023 - July 2023
- Teaching Assistant for Pol. Eco. Workshop at Gokhale Institute of Politics and Economics (GIPE), Pune 09-11 June 2022
- Teaching Assistant for Econometric Method 1 January 2022 - July 2022

# Professional Activities

**Journal Referee**

- Economic Modelling

**Other Professional Activities**

- Discussant at Research Scholars' Day 2026, Department of Economic Sciences, Indian Institute of Technology Kanpur. 13-14 February 2026
- Discussant at the XXXI Economics of Education Association meeting hosted by Santiago de Compostela University. 29-30 June 2023
- Managed the Planning and Policy Research Unit (PPRU), a research center within the Economics and Planning Unit. January 2023 - September 2025
- Association for Mentoring and Inclusion in Economics (AMIE) Mentee. July 2022 - June 2023

- Coordinate weekly seminars at Economics and Planning Unit (with Prof. Debasis Mishra, Prof. E. Somanathan, and Vilok). June 2022 - July 2024

## Software Skills

STATA, LaTeX, ArcGIS, Python and R.

## References

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